Competitive Exam Practice 3 — Answer Key

IEEE 754 Floating-Point Representation and Arithmetic

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Q1 (A). -14.25: magnitude $14.25 = 1110.01_2 = 1.11001_2 \times 2^3$. Sign bit $= 1$ (negative). Exponent $= 3 + 127 = 130 = 1000\,0010_2$. Mantissa $= 1100100\ldots$ (to 23 bits). Assembled as sign, exponent, mantissa, the word is 1 10000010 11001000000000000000000 = 0xC1640000.

Q2 (D). Sign $= 0$. Exponent $= 01111100_2 = 124$, so $E_{\text{true}} = 124 - 127 = -3$. Mantissa $= 101101\ldots$, value $= 1.101101_2 \times 2^{-3}$. Summing the set bits: $2^{-3} + 2^{-4} + 2^{-6} + 2^{-7} + 2^{-9} = 0.125 + 0.0625 + 0.015625 + 0.0078125 + 0.001953125 = 0.21289\ldots \approx 2.13 \times 10^{-1}$. The closest option is 2.27×10^{-1} (D).

Q3 (C). Decode both.

- P = 0xC1800000: sign 1, exponent $10000011_2 = 131$, $E_{\text{true}} = 4$, mantissa $0 \Rightarrow P = -1.0 \times 2^4 = -16$.
- Q = 0x3F5C2EF4: sign 0, exponent $01111110_2 = 126$, $E_{\text{true}} = -1$, mantissa $1.1011100\ldots_2$ $\Rightarrow Q = 1.10111\ldots \times 2^{-1}$.

Product sign $= 1$ (negative). Stored exponent $= E_P + E_Q + 127 = 4 + (-1) + 127 = 130 = 1000\,0010_2$. Mantissa product $1.0 \times 1.10111\ldots = 1.10111\ldots$ (already in $[1, 2)$, no renormalization), so the mantissa field is Q's mantissa unchanged. Assembling sign, exponent, mantissa gives 0xC15C2EF4.

Q4 (D). Exponent all zeros and mantissa all zeros is the **special value** $+0$ — it is not a normalized number (a normalized number requires a nonzero exponent and an implicit leading 1). `0x00000000` is exactly this pattern.

Q5 (B). The bias makes the stored exponent an unsigned field ordered so that all-zeros is the smallest and all-ones the largest exponent, letting ordinary unsigned comparison order floats correctly. (Two's complement *can* represent negative exponents — that is not the reason; the issue is comparison ease.)

Q6 (C). 0.1 is a repeating binary fraction, so each stored value is an approximation; ten additions accumulate rounding error, so x ends up slightly off 1.0 and the equality test may fail. (A) is false because 10×0.1 is not exact in binary; (B) is false because the error direction is not guaranteed; (D) misidentifies the failure mode — no overflow is involved.